

'Design and Development of Belt Bag with Voice-Guided Navigation and Hazard Avoidance Sensors for Visually Impaired Individuals'

Research Description

This study aims to design and develop a wearable belt bag integrated with ultrasonic sensors for obstacle detection to assist visually impaired individuals in navigating different environments safely and independently. The device will provide real-time feedback through voice guidance, vibration, and buzzer alerts whenever an obstacle is detected, allowing users to respond promptly and avoid potential hazards.

The prototype will utilize three ultrasonic sensors strategically positioned at the front, left, and right sides of the belt bag to provide approximately 180° obstacle detection coverage with a maximum detection range of 13 feet. This sensor arrangement enables the system to detect obstacles from multiple directions, enhancing the user's spatial awareness while walking.

The vibration module, buzzer, and voice guidance system will activate only when the ultrasonic sensors detect an obstacle within the specified range. This real-time warning mechanism is designed to improve user safety, mobility, and confidence while minimizing unnecessary alerts.

The voice guidance system will provide clear audio instructions such as "Obstacle ahead on your left in three steps," "Obstacle ahead on your right in two steps," or "Obstacle ahead in one step." It will estimate how many steps away the obstacle is and give voice commands to help the user avoid it, such as "Move to the left," "Move to the right," or "Continue straight." These instructions are intended to help the user avoid obstacles and move safely and independently.

Materials to be used:

1. Breadboard
2. Jumper Wires
3. HCSR04 Ultrasonic sensor
4. Tactile Switch
5. Vibration Module
6. DFR Robot Player
7. Buzzer
8. SD Card
9. ESP32